AL SAFA 2 PARK

Dubai Municipality — AI-Powered Park Design Challenge

DELIVERABLE 7

Material & Landscape Palette

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Written specification. Visual mood-board renders of this palette are produced separately (Gemini/Nano Banana) using this document as the brief.

7.1 Guiding Principles

- **Light over dark.** Dark hardscape surfaces routinely reach temperatures that burn bare skin in Dubai summers. Every primary paved surface in this palette is light-toned (high solar reflectance) by default, not as a stylistic choice.
- **Regionally proven planting, not generic "desert landscaping."** Every tree species below is one Dubai Municipality already plants at scale in its own projects — not a species chosen from general drought-tolerant-plant knowledge.
- **Water honesty.** Where a material or planting choice trades climate comfort for water demand (turf is the clearest case), that tension is stated directly rather than hidden — it belongs in the Sustainability Strategy (Deliverable 9) conversation.
- **Zone-specific, not uniform.** Planting and materials follow the same five confirmed zones from Deliverable 2 — each has a different environmental job, so each gets a different palette.

7.2 Planting Palette by Zone

Zone	Trees / large shrubs	Understory / groundcover	Rationale
Buffer Grove (NW)	Ghaf (Prosopis cineraria) — UAE national tree; Samar (Acacia tortilis)	Low windbreak shrub layer, gravel mulch (light-toned, not dark)	Both species are Dubai Municipality's own windbreak/desert planting choices — deep-rooted, built for exactly this edge's confirmed harsh sun+wind exposure
Shaded Promenade (NE)	Neem (Azadirachta indica)	Low groundcover between canopy gaps	DM's own roadside shade tree of choice — fast dappled canopy, heat/drought/salt tolerant, near disease-free in Dubai conditions
Arrival Commons (SE)	Sidr (Ziziphus spina-christi) or Neem for entrance shade	Bougainvillea, Tecoma, Plumbago (flowering, low-water)	A more ornamental, welcoming character for the entrance sequence, still fully drought-adapted — not water-hungry ornamentals
Central Green Commons	Shade trees framing Playground and Picnic Grove (Ghaf / Neem mix)	Irrigated turf for the Event Lawn — see water flag below	Balances the brief's open-lawn requirement against real water cost
Active Edge (SW) — Quiet Turf Buffer & Transition Zone	Sidr for shade at the transition structures	Lantana (low, hardy, non-obstructive near courts for sightlines/security)	Kept deliberately low near the courts and mosque-facing buffer — visibility and quiet character both matter here

[VERIFIED] Ghaf, Samar (Acacia tortilis), Sidr, and Neem are all confirmed as species Dubai Municipality is actively planting in its own 2024-2025 landscaping projects (Sheikh Zayed Road, Al Khail Road, and multiple residential districts) — sourced directly, not inferred from general arid-climate planting guides.

[HONEST FLAG] The Central Green Commons' Event Lawn trades water for climate performance: irrigated grass is the best-performing surface for daytime microclimate cooling of any material considered here, but it is also the highest water-demand choice in this palette. Dubai Municipality's own approach — irrigation drawn entirely from recycled/treated water rather than potable supply — is the model to carry into Deliverable 9 rather than treating this as an unresolved contradiction.

7.3 Hardscape Materials

Surface	Material	Why
Primary paths, plazas, promenade	Light-toned permeable interlocking concrete pavers (PICP) or light natural stone (travertine/limestone)	High solar reflectance keeps surface temperature down for bare feet and mobility aids; permeability supports the site's soft-landscape water strategy. Dark asphalt/dark pavers explicitly avoided.
Playground safety surfacing	EPDM poured-in-place rubber, light tone specified	Matches the equipment heights already sourced in Deliverable 5 (1.2m toddler / 2.4m junior) via EN 1177 critical-fall-height depth calculation — light tone specifically reduces surface temperature versus standard dark EPDM
Court surfacing (Padel / Basketball)	Synthetic sports surface (acrylic or turf-infill, heat-mitigated product line)	Standard for both sports; flagged below re: heat performance
Perimeter / buffer planting beds	Light gravel mulch, not dark gravel or pebbles	Dark gravel/pebbles cause glare and heat discomfort at ground level — avoided per current UAE landscape practice
Event Lawn	Irrigated natural turf, not artificial	Artificial turf reaches 60–80°C in direct Dubai sun — unusable and unsafe for the open, family-use function this lawn is meant to serve

7.4 Two Findings That Affect Earlier Deliverables

[COMPLIANCE GAP — NEEDS ACTION] Dubai Municipality requires 70–80% shade coverage over active play areas as a stated compliance minimum, not a design preference. The current Deliverable 4 masterplan places shade canopies along the Shaded Promenade but does not confirm a specific shade-coverage percentage over the Toddler or Junior Playground footprints themselves. This needs to be checked geometrically against the actual playground polygons before Deliverable 6 visualization — if coverage falls short, additional canopy placement directly over the playground zones (not just the adjacent promenade) will be needed.

[STANDARD CORRECTION] Deliverables 4 and 5 cited the CPSC Public Playground Safety Handbook (a US standard) for playground age-separation and equipment-height practice. The standard actually recognized by Dubai Municipality, Trakhees, and KHDA is EN 1176 (equipment) and EN 1177 (safety surfacing) — the European standard used throughout the UAE. The underlying age-separation logic and the 1.2m/2.4m height figures already used remain reasonable in practice, but the Complete Design Report (Deliverable 11) should cite EN 1176/1177 as the operative regional standard rather than CPSC.

[OPEN QUESTION — NOT RESOLVED HERE] Neither the Basketball nor Padel court has a shade structure in the current masterplan (official FIBA/FIP don't require one, and Deliverable 4's space budget is already tight). Given Dubai's climate, this is worth a deliberate decision rather than a default omission — even partial shade over spectator/seating areas at the courts, if not the full court, is worth considering. Flagged for Deliverable 4 revision or explicit acceptance as a scoping trade-off, not decided unilaterally here.

7.5 Handoff to Visualization

This document, together with the confirmed masterplan (Deliverable 4) and section/elevation heights (Deliverable 5), is the grounding brief for Deliverable 6's visual outputs — the actual rendered mood board, material swatches, and 3D/spatial visualizations are produced in Gemini/Nano Banana, not here, consistent with this project's tool division. Handing over real, sourced species and material names (Ghaf, Sidr, Neem, light PICP paving, EPDM light-tone surfacing) rather than generic "desert park" prompting should materially improve render accuracy.

Sources: Deliverables 1–5 (this project); Dubai Municipality public landscaping project records (2024–2025); EN 1176 / EN 1177 (European playground equipment and surfacing standards, UAE-recognized); UAE landscape and hardscape material practice (multiple current industry sources). Full AI-tool-by-tool methodology log: Deliverable 10 — AI Methodology Report.